

Derivation of the Probability Density Function for the Laplace Distribution

To derive the Probability Density Function (PDF) for the Laplace Distribution, we begin with the concept of the exponential distribution, whose probability density function is given as:

$$f(y; \lambda) = \lambda e^{-\lambda y}; y > 0, \lambda > 0.$$

Now, let's consider $X = |Y_1 - Y_2|$, where Y_1 and Y_2 each follows an exponential distribution with parameter $\lambda = \frac{1}{\beta}$. That is, the distribution of X is the difference of two independent, identically distributed (i.i.d.) exponential random variables with rate parameter $\lambda = \frac{1}{\beta}$.

To find the PDF of X , we'll use the cumulative distribution function (CDF) method:

$$F_X(x) = P(X \leq x) = P(|Y_1 - Y_2| \leq x)$$

$$= P(-x \leq (Y_1 - Y_2) \leq x) \quad [X \text{ ranges from } -\infty \text{ to } \infty]$$

$$= F_Y(x) - F_Y(-x).$$

where $F_Y(\cdot)$ is the CDF of the exponential distribution. The derivative of $F_X(x)$ with respect to x will give the PDF of X , which is the Laplace distribution. Thus, Laplace distribution is the distribution of the difference of two independent identically distributed exponential random variables centered at a value, i.e. here it is 0.

$$\begin{aligned} f_X(x) &= \frac{d}{dx} (F_Y(x) - F_Y(-x)) \\ &= \frac{d}{dx} \left[\left(1 - e^{-\lambda(x)} \right) - \left(1 - e^{-\lambda(-x)} \right) \right] \\ &= \frac{d}{dx} \left[-e^{-\lambda(x)} + e^{-\lambda(-x)} \right] \\ &= \frac{d}{dx} \left[e^{\lambda x} - e^{-\lambda x} \right] \\ &= \left[\lambda e^{\lambda x} + \lambda e^{-\lambda x} \right] \\ &= \lambda \left[e^{-\lambda(-x)} + e^{-\lambda x} \right] \\ &= \frac{\lambda}{2} e^{-\lambda|x|}; -\infty < x < \infty. \end{aligned}$$

i.e. $f_X(x) = \frac{\lambda}{2} e^{-\lambda|x|}$; $-\infty < x < \infty$. The absolute value function $|x|$ in the exponent ensures that the density function, $f_X(x)$, is symmetric about 0.

Finally, replacing λ with $\frac{1}{\beta}$, and shifting the distribution to be centered at μ , we define the desired density function as

$$f_X(x; \mu, \beta) = \frac{1}{2\beta} e^{-\frac{|x-\mu|}{\beta}}; -\infty < x < \infty.$$

The Laplace distribution, also known as the double exponential distribution, arises naturally from the difference of two exponential distributions and provides a flexible model for various applications, especially when dealing with symmetric data with a sharp peak at the mean and heavier tails compared to the normal distribution.

If $\mu=0$ and $\beta=1$, then the above function reduces to

$$f_X(x) = \frac{1}{2} e^{-|x|}; -\infty < x < \infty.$$

This is the standard form of double exponential distribution. It is also known as Poisson's first law of error.

Historical Remarks:

The double exponential distribution was introduced by Laplace in 1774 as the probability distribution for which the likelihood function is maximized by setting the location parameter equal to the median of an odd number of independent, identically distributed observations. The

distribution defined $f_X(x; \mu, \beta) = \frac{1}{2\beta} e^{-\frac{|x-\mu|}{\beta}}$; $-\infty < x < \infty$ is known as the first law of Laplace

because Laplace initially proposed it for modeling errors. The well-known normal distribution is known as second law of Laplace, as Laplace later derived it as an approximation for sums of random errors. The distribution is known under different names. The most common one is Laplace double exponential distribution. Feller called it bilateral exponential. Weida called the

standard double exponential distribution $f_X(x) = \frac{1}{2} e^{-|x|}$; $-\infty < x < \infty$. as Poisson's first law of error.